

ADDISON W. MOTTO

awm82@cornell.edu | 609-216-4166 | cornell-mae-ug.github.io/AddisonMotto

SUMMARY

Mechanical engineer experienced across the full hardware lifecycle — CAD, GD&T drawings, prototyping, and test-rig validation — seeking to bring a rigorous design-build-test approach to challenging mechanical engineering problems.

ACADEMICS

Cornell University

2022 — Present

M.Eng. Systems Engineering | Jan 2026 — Dec 2026

B.S. Mechanical Engineering | 2022 — 2026 | GPA: 3.83 | Magna Cum Laude | Dean's List (All Semesters)

TECHNICAL SKILLS

SolidWorks (CAD + Simulation/FEA) · Onshape · Ansys (Mechanical, Fluent & Thermal) · GD&T (ANSI Y14.5) · DFM/DFA · Autodesk Inventor · Fusion 360 · CREO CAD · MATLAB · Python

EXPERIENCE

Helbling Laboratory, Cornell University — Graduate Researcher

2026 — Present

- Designed and experimentally validated a water-braking mechanism for Gammabot, a sub-1-gram flapping-wing robot, characterizing drag performance across multiple brake configurations.
- Fabricated and assembled millimeter-scale robot components using carbon-fiber processing and precision manufacturing.
- Engineered a pulley/magnetic-rail test setup with high-speed imaging to extract braking kinematics.

Creative Engineering LLC — Mechanical Design Intern

Summer 2025

- Delivered a fully functional 5-ft interactive wind turbine exhibit on schedule, as measured by successful client sign-off, by owning CAD modeling, vendor coordination, and final on-site assembly.
- Designed components for 6 client products — including CAD modeling for the Times Square New Year's Eve Ball — applying DFM/DFA and GD&T across machined, welded, 3D-printed, laser-cut, and injection-molded parts.
- Performed statics and materials-science calculations to support the design of patent-pending devices, manufacturing equipment, and mechanical tooling.
- Presented CAD models and technical rationale to senior engineers and clients in recurring design reviews.

Raytheon (RTX Corp) — Mechanical Engineering Intern

Summer 2024

- Performed rapid design iteration within Raytheon's Rapid Hardware Solutions team, accelerating hardware build cycles for advanced radar and sensor systems.
- Selected as 1 of ~10 summer interns to present technical work to engineering leadership, showcasing preset cable-cutting programs that cut assembly time by 1–2% across a 20-technician harness shop and remained in use post-internship.
- Interpreted engineering drawings and identified GD&T inconsistencies against test procedures, flagging and correcting dozens of issues before production.
- Performed systematic root-cause analysis on product failures, resolving issues under strict DoD test protocols.

Hikma Pharmaceuticals — Manufacturing Intern

Summer 2023

- Supported operations under strict SOPs and cGMP protocols at a sterile injectables facility, gaining hands-on exposure to regulated, high-volume pharmaceutical manufacturing in full compliance with FDA/DEA standards.
- Supported uninterrupted, high-speed production, as measured by consistent line uptime, by assisting operation of automated, high-volume liquid-filling lines.

CORNELL HIGHLIGHTS

Cornell Electric Vehicles — Student Project Team

2023 — 2025

- Designed power-transmission, steering, and braking systems integrating motors and sensors, enabling competition-ready performance at the Shell Eco-Marathon.
- Delivered a custom load-cell/tachometer-based dynamometer, as measured by on-time delivery under an aggressive timeline, by designing and fabricating the system primarily from salvaged parts.

Cornell Engineering Learning Initiatives — Tutor

2023 — 2024

- Selected by Cornell's Engineering Learning Initiatives to tutor peers in Physics, Multivariable Calculus, and Chemistry.

Interests: Piano · Skiing · Running · Fishing · Travel